



Noetherian sets and rings in nondeterministic mathematics

Constructive and Synthetic Methods in
Algebra, Geometry, and Foundations of Mathematics
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partially joint work with



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A brittle definition

A set X is **streamless** iff for every function $f : \mathbb{N} \rightarrow X$, there are numbers $i < j$ with $f(i) = f(j)$.

- Classically, streamless $\iff$ finite.
- Constructively, (finitely enumerable $\vee$ subfinite) $\implies$ streamless $\implies \neg$ Dedekind-infinite.

Recall: A set is ...

- finitely enumerable iff it is a quotient of a set of the form $[n] = \{k \in \mathbb{N} \mid k < n\}$,
- subfinite iff it embeds into a set of the form $[n]$,
- Dedekind-infinite iff $\mathbb{N}$ embeds into it.

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A question

Is the cartesian product of streamless sets streamless? [Richman–Stolzenberg 1993]

A quasi-order X is **well** iff for every function $f : \mathbb{N} \rightarrow X$, there are numbers $i < j$ with $f(i) \leq f(j)$.

Is the cartesian product of well quasi-orders well?

In this talk

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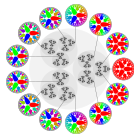
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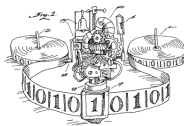
Formal topology

1



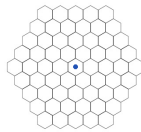
Growing a counterexample

2



Stateful realizability

3



Unique choice

4



Perspectives

5

What is a function in constructive mathematics?

Function types are brittle.

(naive) statement	classically / constructively / recursively			
0 Every function is computable.	✗	?	✓	
1 Every function has a zero or not.	✓	?	✗	
2 Every path leaves the Kleene tree.	✗	?	✓	
3 There is a surjection $\mathbb{N} \twoheadrightarrow \mathbb{R}$.	✗	?	✗	True in parametrized realizability.
4 The continuum hypothesis holds.	?	?	✗	

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With **adverse effects** throughout mathematics:

5 Products of wf. relations are wf.	✓	?		
6 Minimum principle for unif. cont. functions.	✓	?	✗	“For $f : [0, 1] \rightarrow \mathbb{R}_{>0}$, there is $c > 0$ with $f \geq c$.”
7 Every field has an algebraic closure.	✓	?	✗	Implies a form of choice.
8 Hilbert’s basis theorem.	✓	?		“If A is Noetherian, so is $A[X]$.”
9 Locally Noetherian implies Noetherian.	✓	?		“If all $A[f_i^{-1}]$ are Noetherian, so is A .”
10 Krull’s lemma.	✓	?		“If $x \in \bigcap_{\mathfrak{p}} \mathfrak{p}$, then x is nilpotent.”
11 Fundamental theorem of Galois theory.	✓	?		“If $x \in L^{\text{Aut}_K(L)}$, then $x \in K$.”

The formal space of functions



Instead of referencing the under-specified **collection of functions** $\mathbb{N} \rightarrow X$, consider the explicit **formal space of functions** $\mathbb{N} \rightarrow X$.

A brittle definition

A set X is **streamless** iff the **collection of functions** $X^{\mathbb{N}}$ can be written as a **union** as follows:

$$X^{\mathbb{N}} = \bigcup_{i < j} \{f \mid f(i) = f(j)\}.$$

A robust definition

A set X is **Noetherian** iff the **formal space** $X^{\mathbb{N}}$ is **covered** as follows:

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That is, a set X is Noetherian iff equivalently ...

- 1 there is a **dialogue tree** arguing, just from querying a **black box** $f : \mathbb{N} \rightarrow X$, that there exist numbers $i < j$ with $f(i) = f(j)$;
- 2 **everywhere**, for every function $f : \mathbb{N} \rightarrow X$ there exist numbers $i < j$ with $f(i) = f(j)$;
- 3 Dup bars the empty sequence, where $\text{Dup } x_0 \dots x_{n-1} := (\exists i < j. x_i = x_j)$.

✓ The cartesian product of Noetherian sets is Noetherian, **without foundational assumptions**.

Growing a counterexample

We seek two equivalence relations R and S on $\mathbb{N}$ such that:

- 1 Every function $f : \mathbb{N} \rightarrow \mathbb{N}$ is R -good, i. e. there exist numbers $i < j$ such that $f(i) R f(j)$.
- 2 Every function $f : \mathbb{N} \rightarrow \mathbb{N}$ is S -good.
- 3 The relations R and S are **orthogonal** in that $x R y \wedge x S y \Rightarrow x = y$.
- 4 Every function $\mathbb{N} \rightarrow \mathbb{N}/R$ lifts to a function $\mathbb{N} \rightarrow \mathbb{N}$, as if countable choice was available, and analogously for S .

Then:

- The quasi-orders $(\mathbb{N}, R)$ and $(\mathbb{N}, S)$ are well, but their product is not.
- The sets $\mathbb{N}/R$ and $\mathbb{N}/S$ are streamless, but their product is not.

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A fantasy

Realize R and S as **dynamic** entities never to be completed. Starting empty, add more identifications to R and S **step by step**—but **only when challenged**.

Key lemma: Let $f : \mathbb{N} \rightarrow \mathbb{N}$ be a function. Then f is $=$ -good or there are two values which exceed all numbers **currently** related by R or by S .

Recall the **Brouwer–Heyting–Kolmogorov interpretation**:

- A witness of $P \wedge Q$ is a pair consisting of a witness of P and a witness of Q .
- A witness of $P \Rightarrow Q$ is a **construction** which converts witnesses of P into witnesses of Q .
- A witness of $\forall x \in \mathbb{N}. P(x)$ is a **construction** which converts numbers x into witnesses of $P(x)$.
- A witness of $\exists x \in \mathbb{N}. P(x)$ is a pair consisting of a number x and a witness of $P(x)$.

But what is a **construction**?

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But what is a **construction**?

Is there a witness of the **axiom of choice**?

$$(\forall x \in A. \exists y \in B. R(x, y)) \implies (\exists f : A \rightarrow B. \forall x \in A. R(x, f(x)))$$

Can every **construction** which converts x 's into y 's be converted into a suitable *function* $A \rightarrow B$?

Stateful realizability

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Kleene realizability [Kleene 1945]

Use **Turing machines** as “constructions”.

A machine e realizes P , denoted $e \Vdash P$, iff ...

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Stateful realizability [Cohen–Faro–Tate 2019]

Use (a specified flavor of) **stateful nondeterministic Turing machines** as “constructions”.

A machine e realizes P in state s , denoted $s \mid e \Vdash P$, iff ...

Here: States are pairs (R, S) of finite lists of natural numbers such that R^* and S^* are orthogonal, allowed to be extended (but not shortened).

Unique choice

In the stateful realizability tripos

In a state s , a function $\mathbb{N} \rightarrow \mathbb{N}$ is

- 1 a metatheoretic function $f : \mathbb{N} \rightarrow \mathbb{N}$

which is **tracked** by a machine e : For every state $s' \geq s$ and for every number $n \in \mathbb{N}$, the application $e \cdot n$ should terminate with the numeral for $f(n)$.

Such kinds of functions are R -good and S -good by design.

In the stateful realizability topos

In a state s , a function $\mathbb{N} \rightarrow \mathbb{N}$ is

- 1 a relation F , denoted $s' \mid e \Vdash F(x, y)$ (in state s' , machine e is a realizer of x being mapped to y) together with
- 2 a machine t which realizes totality: $s \mid t \Vdash \forall x \in \mathbb{N}. \exists y \in \mathbb{N}. F(x, y)$, and
- 3 a machine u which realizes single-valuedness:
 $s \mid u \Vdash \forall x \in \mathbb{N}. \forall y, y' \in \mathbb{N}. F(x, y) \wedge F(x, y') \Rightarrow y = y'$.

Our particular model validates unique choice for **nontrivial combinatorial reasons**.

A set X is **Noetherian** iff the **generic** function $\mathbb{N} \rightarrow X$ has duplicate values.

A robust definition

A ring A is **Noetherian** iff for the **generic** function $f : \mathbb{N} \rightarrow A$, there exists a number $n \in \mathbb{N}$ such that $f(n) \in (f(0), \dots, f(n-1))$.

Then:

- 1 **Without** any decidability or coherence assumptions:
If A is Noetherian so is $A[X]$. [Coquand–Persson 2001]
By the classical proof! [B. 2026]
- 2 Let $1 = f_1 + \dots + f_n$ in A . Then A is Noetherian iff all localizations $A[f_i^{-1}]$ are. [B. 2026]
- 3 A ring A is Noetherian iff $\mathcal{O}_{\text{Spec } A}$ is. [B. 2026]

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For almost no metric space X does the **generic** (modulated) Cauchy sequence $\mathbb{N} \rightarrow X$ converge.